



HSE University, Faculty of
Computer Science
Master's Programme in Data
Science

2026

Economic and Ecological Losses from Wildfires using Machine Learning

Master's Thesis Defense

Muhammad Suliman

Supervisor: Dr. Ramon Antonio Rodrigues Zalipynis



Presentation Plan

1. Motivation and research gap
2. Data foundation and methodology
3. Modeling and validation results
4. Risk-assessment and ecological screening outputs
5. Supporting work, limitations, and conclusion
6. References and appendix evidence

Source: Thesis structure: Introduction, Methodology, Results, Discussion, Conclusion, and Appendices.



Motivation: From Fire Prediction to Consequence Assessment

The thesis focuses on losses, not only whether or where fires occur.

Core problem

- Wildfire ML often predicts occurrence, susceptibility, spread, or burned area.
- Defense decisions require consequence-oriented outputs.
- This thesis estimates monetary wildfire risk and adds ecological exposure screening.

Research direction

- Public data and environmental predictors.
- Geographic validation and interpretable outputs.

Defense focus

Risk prediction is useful only when it can be translated into interpretable, loss-oriented outputs.

Thesis direction

From fire prediction to wildfire-loss assessment.

Source: Thesis Abstract, Chapter 1, and Chapter 2.



Research Gap, Novelty, and Contribution

The thesis proposes a general consequence-oriented wildfire-loss framework.

Gap in previous work

- Many wildfire ML studies remain hazard-centric.
- Economic and ecological consequences are often analysed separately.
- Random validation may overestimate geographic transferability.

New contribution

- County-level FEMA/NRI wildfire EAL modeling pipeline.
- Environmental augmentation using terrain, land cover and climate.
- State-wise grouped validation and ecological screening layer.

Key insight

Loss-oriented wildfire assessment improves when environmental hazard context is modeled together with exposure, vulnerability and resilience.

Source: Thesis Chapter 1: novelty and contribution framing.



Literature Review Synthesis

The review connects wildfire prediction, loss assessment, ecological vulnerability and emerging AI.

Main literature streams

- Wildfire occurrence, susceptibility, detection and spread modeling.
- Economic loss, asset exposure, suppression cost and risk management.
- Ecological vulnerability, ecosystem exposure and post-fire recovery.
- Emerging AI, LLM and VLM tools for wildfire decision support.

Takeaway for this thesis

The literature supports moving beyond hazard-only prediction toward integrated consequence-oriented wildfire-loss assessment.

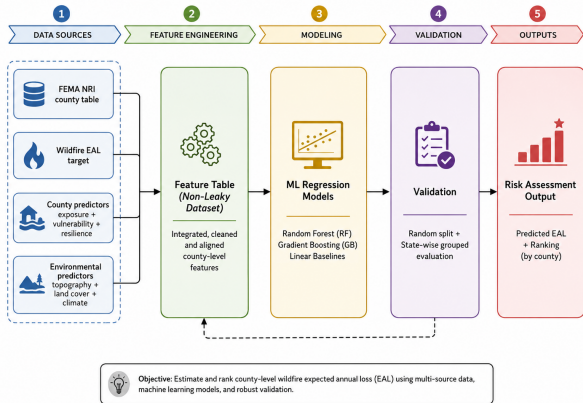
How it shaped this study

- FEMA NRI EAL supplies a monetary target.
- Environmental variables add wildfire-relevant context.
- Ecological screening provides an exploratory proxy layer.



Methodology Pipeline

Public data → features → ML models → validation → risk outputs.



FEMA/NRI target and public predictors → model training → geographic validation → monetary and ecological screening outputs

Source: Thesis Chapter 3, Figure 3.1.



Data Foundation

The model combines public loss indicators with county predictors and environmental context.

Target and county-level predictors

- Target: wildfire Expected Annual Loss (EAL) from FEMA NRI.
- Exposure/value: population, building value, agriculture value.
- Social capacity: social vulnerability and community resilience.
- Derived densities: per-area and per-capita measures.

Environmental augmentation

- Topography: elevation and slope.
- Land cover: forest, shrub, grass, developed fractions.
- Climate: warm-season temperature, precipitation, wind, and VPD.

Source: Thesis Chapter 3, Sections 3.4-3.7.



Target Variable and Log Transformation

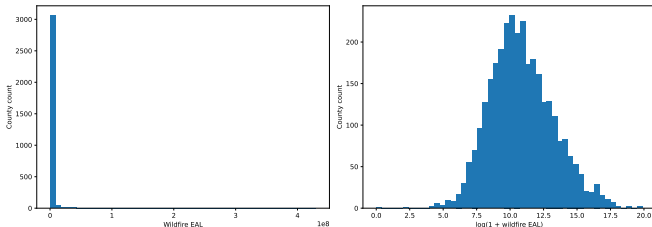
The raw wildfire EAL target is strongly right-skewed.

Modeling target

- County-level wildfire EAL is used as the monetary loss-oriented target.
- The target is transformed as:

$$y = \log(1 + \text{wildfire_eal})$$

- The log target reduces skewness and stabilizes the regression task.



Source: Thesis Chapter 4, Section 4.2 and Figure 4.1.



Modeling and Validation Strategy

The thesis evaluates both predictive performance and geographic transferability.

1. Construct a cleaned non-leaky modeling dataset.
2. Train baseline models using county-level exposure, vulnerability, and resilience predictors.
3. Add environmental predictors and compare against the baseline.
4. Evaluate models using both random splitting and state-wise grouped validation.

Grouped validation

State-wise grouped validation holds out entire states from training, giving a stricter test of spatial generalization than random splitting.

Source: Thesis Chapter 3, modeling workflow and validation strategy.



Baseline County-Level Model

Public county-level indicators already contain meaningful predictive signal.

Baseline model comparison: 5-fold CV
 R^2

Model	CV R^2
Linear Regression	0.318
Ridge Regression	0.321
Gradient Boosting	0.442
Random Forest	0.504

Best baseline: Random Forest

Test R^2	Test RMSE	CV R^2
0.483	1.675	0.504

Main reading

- Random Forest provides the strongest baseline result.
- Exposure, value, vulnerability and resilience variables already carry predictive signal.
- This establishes the benchmark before environmental augmentation.

Caution

Baseline performance is not final evidence of geographic transferability; this is tested later using state-wise grouped validation.

Source: Thesis Chapter 4, Section 4.3 and Table 4.1.



Environmental Augmentation of the County Model

Topography, land cover and climate substantially improve predictive performance.

Random Forest comparison

Feature set	CV R^2
County-only baseline	0.520
Environmental only	0.769
Combined model	0.795

Best-performing feature set:
Combined model

Test R^2	RMSE	CV R^2
0.829	1.011	0.795

Why performance improves

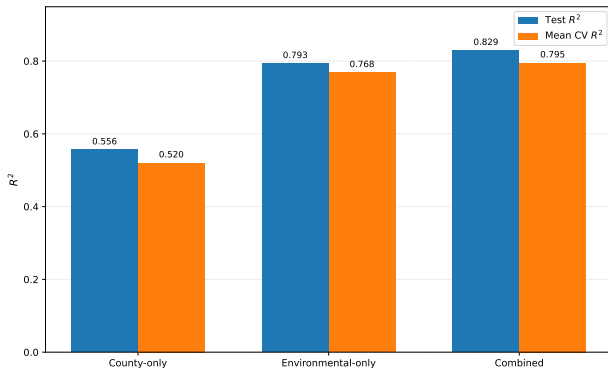
- Terrain, land cover and climate provide wildfire-relevant hazard context.
- Environmental-only predictors already contain strong predictive signal.
- Combining hazard context with exposure and vulnerability gives the strongest result.

Source: Thesis Chapter 4, Section 4.6 and Table 4.3.



Performance Gain from Environmental Predictors

The combined model gives the strongest performance under random-split evaluation.



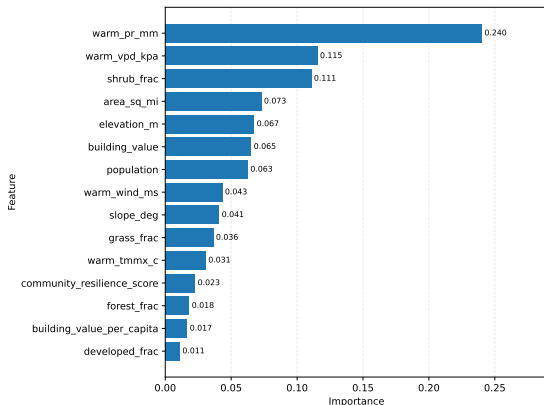
Key takeaway

Adding environmental predictors improves the Random Forest model substantially, and the combined feature set gives the strongest overall performance.



Feature Importance in the Combined Model

Climate and exposure variables dominate the combined Random Forest model.



Main insight

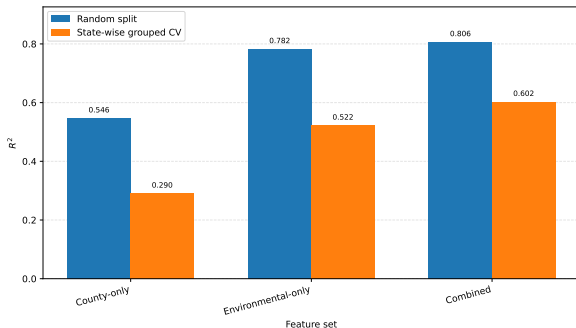
- Warm-season precipitation is the strongest predictor.
- Climate and land-cover variables add information beyond county exposure and value variables.
- This supports the thesis claim that environmental augmentation matters.

Source: Thesis Chapter 4, Section 4.7 and Figure 4.8.



State-Wise Geographic Validation

The model remains useful under stricter spatial generalization.



Grouped-validation result

- Performance decreases compared with random split, as expected.
- The combined model remains the best setup.
- Combined grouped $R^2 \approx 0.603$.

Defense relevance

This evaluation avoids overly optimistic random mixing and makes the thesis more robust.

Source: Thesis Chapter 4, Section 4.8 and Figure 4.9.



From Model Score to Risk-Assessment Output

The pipeline produces interpretable county-level risk outputs, not only model metrics.

Outputs

- Predicted wildfire EAL as a monetary estimate.
- Risk ranking for county prioritization.
- Error analysis for under- and overprediction.
- Ecological screening as a first overlap signal.

$$\widehat{\text{EAL}} = \exp(\hat{y}) - 1$$

Rank	County	Predicted EAL
1	Riverside, CA	\$146.9M
2	San Diego, CA	\$127.4M
3	Los Angeles, CA	\$123.3M
4	San Bernardino, CA	\$75.3M
5	Orange, CA	\$73.9M

Source: Thesis Chapter 4, Section 4.9 and Table 4.6.



Exploratory Ecological Risk Proxy

The ecological layer is a screening proxy; it does not estimate measured ecological loss.

Proxy construction

Component	Meaning
Natural vegetation fraction	Forest + shrub + grass land-cover fractions
Predicted risk percentile	Percentile rank of predicted wildfire EAL
Ecological risk proxy	Overlap of monetary wildfire risk and natural vegetation exposure

Proxy formula

Ecological risk proxy = predicted risk percentile
× natural vegetation fraction

Source: Thesis Chapter 4, Tables 4.7–4.8.

What it captures

- High predicted monetary wildfire risk.
- High natural vegetation exposure.
- Locations suitable for further ecological analysis.

What it does not claim

It is not measured ecological loss; it is an exploratory screening layer for prioritisation and future analysis.



Top Counties by Ecological Risk Proxy

High values identify overlap between predicted monetary wildfire risk and natural vegetation exposure.

Top five counties from the full Top 25 table

County, State	Natural veg.	Predicted EAL	Proxy
Lincoln, NM	0.9904	\$7.66M	97.04
Yavapai, AZ	0.9765	\$20.28M	97.04
Elko, NV	0.9715	\$15.80M	96.23
Coconino, AZ	0.9690	\$20.17M	96.22
Navajo, AZ	0.9762	\$9.33M	95.96

Interpretation

All listed counties are in the **Very High** ecological risk category. The full Top 25 results are reported in the thesis.

How to interpret this output

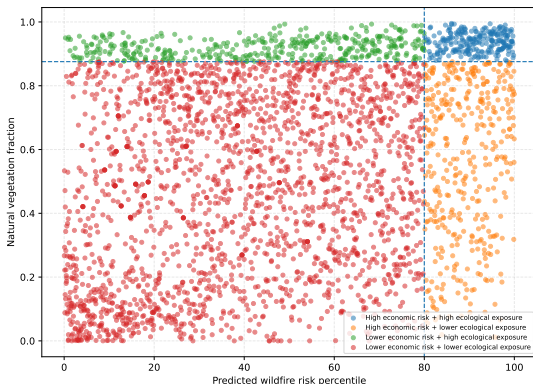
- These counties are candidates for more detailed ecological vulnerability or recovery analysis.
- High proxy values do not mean that measured ecological losses have been observed.
- The output extends monetary EAL prediction into exploratory ecological screening.

Source: Thesis Chapter 4, Table 4.8.



Economic-Ecological Overlap

The output separates high monetary risk from high ecological exposure.



High-high group

- **238 counties** combine high predicted EAL risk and high natural vegetation exposure.
- Mean ecological risk proxy: **85.62**.
- Mean natural vegetation fraction: **0.94**.
- Mean predicted wildfire EAL: **\$3.47M**.

Source: Thesis Chapter 4, Figure 4.15 and Table 4.9.



Summary of Main Experimental Results

The strongest results come from combining county indicators with environmental predictors.

environmental variables substantially improve predictive performance, and the combined model remains the strongest setup even under state-wise validation.

Table 4.12: Summary of county-level wildfire EAL modeling results.

Model setup	Best model	Test R^2	CV / grouped R^2	Notes
County-only baseline	Random Forest	0.483	0.504	Accessible county variables only
Environmental-only (CONUS)	Random Forest	0.793	0.768	Topography, land cover, climate
Combined baseline + environment (CONUS)	Random Forest	0.829	0.795	Standard random cross-validation
Combined baseline + environment (state-wise)	Random Forest	0.806	0.602	Grouped cross-validation by state

Main conclusion from the results

The combined model substantially improves over the county-only baseline under random validation and remains the strongest model under grouped geographic validation.



Supporting Published Work: An Giang, Vietnam

Supporting work; the FEMA/NRI loss-assessment pipeline remains the core contribution.

ults from the supporting An Giang, Vietnam wildfire]

Metric	Value
Overall accuracy	91.4%
Significant fire recall	74.8%
Significant fire precision	92.5%
F1-score	82.7%
Specificity	97.7%
False positive rate	2.3%

Defense relevance

- The supporting Vietnam study was presented at DCHPC 2026.
- The associated conference paper is published in IEEE Xplore.
- It demonstrates the broader wildfire-ML direction, while the thesis focuses on FEMA/NRI monetary loss assessment.

Source: Thesis Chapter 4, Section 4.12; Appendix C; IEEE Xplore document 11517247.



Limitations and Future Work

The thesis establishes a first baseline, not a final complete wildfire-loss framework.

Current limitations

- County-level aggregation does not capture event- or parcel-level damage mechanisms.
- Ecological output is a screening proxy, not measured ecological loss.
- Geographic transferability remains imperfect under grouped validation.

Future extensions

- Add event-level losses, suppression cost, infrastructure, health, and ecological indicators.
- Improve uncertainty quantification and regional calibration.
- Develop the Creek Fire workflow into a fine-scale spatio-temporal model.

Source: Thesis Chapter 5 and Chapter 6.



Conclusion: What the Thesis Demonstrates

Public data can support a meaningful baseline for consequence-oriented wildfire-loss assessment.

1. Loss-oriented framing

Moves beyond hazard prediction toward monetary wildfire EAL estimation and risk ranking.

2. Environmental signal

Topography, land cover, and warm-season climate substantially improve prediction.

3. Broader framework

Adds ecological screening as a first bridge toward consequence-oriented wildfire assessment.

Final defense message

The framework is reproducible and adaptable to other regions when comparable hazard, exposure, vulnerability, environmental, and loss-related data are available.



Thank you for your attention
Questions and Discussion

Muhammad Suliman

Master's Thesis Defense

Economic and Ecological Losses from Wildfires using Machine Learning

Discussion

I welcome your questions and comments.

Repository and supplementary materials are
available via the QR code.

Source: Project repository and supplementary materials.



Project GitHub repository



Selected References: Data and Methodology

Core sources directly supporting the empirical thesis pipeline.

- Federal Emergency Management Agency (FEMA). (2025). *National Risk Index Technical Documentation and Methodology*.
- Abatzoglou, J. T. (2013). Development of gridded surface meteorological data for ecological applications. *International Journal of Climatology*, 33, 121–131. DOI: 10.1002/joc.3413.
- Dewitz, J. (2021). *National Land Cover Database (NLCD) 2019 Products*. U.S. Geological Survey.
- Roberts, D. R., Bahn, V., Ciuti, S., et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40, 913–929.
- Meyer, H., & Pebesma, E. (2021). Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12, 1620–1633.

Source: Selected references from the thesis methodology and bibliography.



Selected References: Wildfire Loss and Ecological Context

Key literature motivating the consequence-oriented framing.

- Rothermel, R. C. (1972). *A Mathematical Model for Predicting Fire Spread in Wildland Fuels*. USDA Forest Service.
- Chuvieco, E., Aguado, I., Yebra, M., et al. (2010). Development of a framework for fire risk assessment using remote sensing and geographic information system technologies. *Ecological Modelling*, 221, 46–58.
- Ager, A. A., Vaillant, N. M., & Finney, M. A. (2010). A comparison of landscape fuel treatment strategies to mitigate wildland fire risk in the urban interface and preserve old forest structure. *Forest Ecology and Management*, 259, 1556–1570.
- Balch, J. K., Bradley, B. A., Abatzoglou, J. T., et al. (2017). Human-started wildfires expand the fire niche across the United States. *PNAS*, 114, 2946–2951.
- Arroqante-Funes, P., Mouillot, F., Moreira, B., et al. (2024). Mapping and assessment of ecological vulnerability to wildfires in Europe. *Fire Ecology*, 20, 98.

Source: Selected references from Thesis Chapter 2 and the ecological-screening motivation.



Appendix A: Supporting Outputs from the FEMA/NRI Pipeline

The appendix preserves additional diagnostic and supporting tables.

The main ecological risk proxy results from Notebook 7 are presented in Chapter 4. This appendix section preserves the supporting component values used to construct the natural vegetation fraction for the top ecological-risk counties. In particular, the table below reports the forest, shrub, and grass fractions that were summed to obtain the natural vegetation fraction used in Table 4.8.

Table A.17: Natural vegetation components for the top ecological-risk counties from Notebook 7.

State	County	FIPS	Forest fraction	Shrub fraction	Grass fraction	Natural vegetation fraction
New Mexico	Lincoln	35027	0.1244	0.5603	0.3057	0.9904
Arizona	Yavapai	04025	0.1579	0.7712	0.0474	0.9765
Nevada	Elko	32007	0.0605	0.6652	0.2458	0.9715
Arizona	Coconino	04005	0.2284	0.6181	0.1226	0.9690
Arizona	Navajo	04017	0.1429	0.7070	0.1263	0.9762
New Mexico	Grant	35017	0.2805	0.4978	0.2037	0.9820
Arizona	Gila	04007	0.3459	0.6019	0.0296	0.9774
New Mexico	San Miguel	35047	0.2502	0.6506	0.0846	0.9853

Purpose of appendix evidence

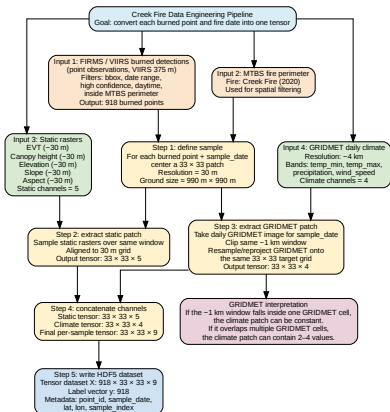
- Preserves supporting outputs from the seven modeling notebooks.
- Documents target construction, missingness, diagnostics, error analysis, ecological component values, and supporting tables.
- Keeps the main presentation focused on the key findings.

Source: Thesis Appendix A.



Appendix B: Creek Fire Workflow

Supporting fine-scale data-engineering prototype for future extension.



Why included

- Demonstrates a fine-scale geospatial data-engineering workflow.
- Uses fire detections, perimeter data, static predictors and lagged climate variables.
- Not the core thesis pipeline because it is not directly linked to monetary EAL.

Source: Thesis Appendix B, Figure B.1.



Appendix C: Research Dissemination

Successful conference presentation of related wildfire work.



Defense relevance

- Certificate is included in Appendix C.
- The An Giang, Vietnam wildfire prediction study was presented at DCHPC 2026.
- The associated conference paper is published in IEEE Xplore.

Scope note

Supporting published work aligned with the thesis direction; the FEMA/NRI pipeline remains the core contribution.

Source: Thesis Appendix C; DCHPC 2026 certificate; IEEE Xplore document 11517247.